Soln: p: x is an even number q: x^2 is an even number.

We shall prove $p \to q$ is True by a direct proof.

$\therefore$ Let p be True. x is an arbitrary even number. By defn 1, $\exists$ an integer K s.t. $x = 2K$.

$\therefore x^2 = (2K)^2 = 4K^2 = 2(2K^2)$. $\because K$ is an integer, $2K^2$ is an integer.

By defn 1, x^2 is an even number. $\boxed{\therefore q \text{ is True}}$ $\square$

4) Show that the additive inverse, or negative, of an even number is an even number using a direct proof.

Soln: p: x is an even number q: $-x$ is an even number

We shall prove $p \to q$ is True by a direct proof.

$\therefore$ Let p be True. x is an arbitrary even number. By defn 1, $\exists$ an integer K s.t. $x = 2K$

$\therefore -x = (-1)\cdot x = (-1)\cdot(2K) = 2K\cdot(-1)$ (Commutative law) $= 2(K\cdot(-1))$ (Associative law) $= 2\cdot(-K)$. $\because K$ is an integer, $\therefore -K$ is an integer.

By defn 1, $-x$ is an even integer. $\boxed{\therefore q \text{ is True}}$ $\square$

5) Prove that if $m+n$ and $n+p$ are even integers, where m, n and p are integers, then $m+p$ is even. What kind of proof did you use?

Soln: p: $m+n$ is an even integer q: $n+p$ is an even integer

p: $m+p$ is an even integer

We shall prove $(p \land q) \to r$ is True by a direct proof.

Let $(p \land q)$ be True. By defn. of AND, p is True, q is True

$\therefore (m+n)$ and $(n+p)$ are both even integers.

By defn. of even, $\exists$ integers K_1 and K_2 such that $(m+n) = 2K_1$ and $(n+p) = 2K_2$

$\therefore (m+n)+(n+p) = 2(K_1+K_2) \Rightarrow m+2n+p = 2(K_1+K_2) \Rightarrow \boxed{m+p = 2(K_1+K_2-n)}$.

$\because K_1, K_2$ and n are integers, (K_1+K_2-n) is also an integer